Questions Extracted from the Lectures (same style as the doctor's bank)

أسئلة مستخرجة من المحاضرات بنفس أسلوب الدكتورة

Source: generated by Claude from the lecture notes in the uploaded archive (Ch1–Ch7, Ch14, Ch16), written in the same style as the doctor's bank: True/False, Multiple choice and Fill-in-the-blank.

- **222** original questions (117 T/F, 68 MCQ, 37 fill-in), grouped per chapter. They are not the doctor's questions; the doctor's bank is a separate file.
- Every answer is taken from the statement in the lecture notes (the reference under each question names the chapter and section). Arabic explanations are short hints.
- The notes cover Chapters 1–7, 14 and 16 only. Chapters 8–13 and 15 were not in the archive, so no questions were generated for them.

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Ch1: Computer Organization — Foundations

True / False — (16) صح / خطأ

1 **A computer architecture describes what a computer should do, while computer organization describes how it works.**

Answer: **True**

Architecture = ماذا يفعل، Organization = كيف يعمل.

المصدر: Ch1 study guide §4

2 **Computer organization is designed first, and computer architecture is started after the organization is finalized.**

Answer: **False**

العكس: المعمارية تُصمَّم أولاً ثم يبدأ التنظيم.

المصدر: Ch1 study guide §4

3 **In the Von Neumann model the program is stored in memory and its instructions are executed sequentially.**

Answer: **True**

من خصائص Von Neumann الأساسية.

المصدر: Ch1 study guide §3

4 **The system bus consists of the control bus, the address bus and the data bus.**

Answer: **True**

system bus = control + address + data.

المصدر: Ch1 study guide §3

5 **The core is a physical piece of silicon that contains one or more processors.**

Answer: **False**

العكس: الـ processor قطعة سيليكون تحتوي core واحداً أو أكثر.

المصدر: Ch1 study guide §6

6 **The IAS computer completed in 1952 introduced the stored program concept attributed to John von Neumann.**

Answer: **True**

IAS: stored program concept، اكتمل 1952.

المصدر: Ch1 study guide §7.1

7 **An IAS instruction word is 40 bits long and contains two 20-bit instructions.**

Answer: **True**

كل تعليمة = 8 بت opcode + 12 بت address.

المصدر: Ch1 study guide §7.1

8 **The IAS Instruction Register (IR) holds the 8-bit opcode of the instruction being executed.**

Answer: **True**

IR يحمل opcode من 8 بتات.

المصدر: Ch1 study guide §7.1

9 **The transistor was invented at Bell Labs in 1958.**

Answer: **False**

الترانزستور 1947؛ الدارة المتكاملة 1958.

المصدر: Ch1 study guide §7.2

10 **Data channels were introduced in second-generation computers such as the IBM 7094.**
Answer: **True**
وحدة I/O مستقلة لها معالج وتعليمات خاصة.
المصدر: Ch1 study guide §7.2

11 **The integrated circuit was invented in 1958 and defines the third generation of computers.**
Answer: **True**
الجيل الثالث = Integrated Circuits.
المصدر: Ch1 study guide §7.3

12 **Moore's Law states that the number of transistors on a chip doubles about every 18 months (since the 1970s).**
Answer: **True**
في السبعينات أصبح التضاعف كل 18 شهراً.
المصدر: Ch1 study guide §7.3

13 **The Intel 8080 was the first chip to contain all CPU components on a single chip.**
Answer: **False**
هذا هو Intel 4004 (1971)؛ 8080 هو أول general-purpose microprocessor.
المصدر: Ch1 study guide §7.4

14 **An embedded system usually has a fixed function that is specific to its application.**
Answer: **True**
بخلاف الحاسوب العام الغرض.
المصدر: Ch1 study guide §8

15 **In SaaS the client manages the operating system and the database.**
Answer: **False**
في SaaS المزوّد (CSP) يدير كل شيء.
المصدر: Ch1 study guide §10

16 **A microcontroller is a single chip containing the processor, ROM, RAM, a clock and an I/O control unit.**
Answer: **True**
computer on a chip.
المصدر: Ch1 study guide §8

Multiple Choice — اختيار من متعدد (10)

1 **Which statement describes computer architecture?**
A. The memory technology used
B. Control signals between components
C. Attributes visible to the programmer ✓
D. Hardware details transparent to the programmer
Answer: **C. Attributes visible to the programmer**
Architecture = visible to the programmer.
المصدر: Ch1 study guide §4

2 **Which of the following is an organizational attribute?**
A. Instruction set
B. Addressing techniques
C. Control signals ✓
D. Number of bits used for data types
Answer: **C. Control signals**
Control signals تفصيل تنظيمي.
المصدر: Ch1 study guide §4

3 **The four main structural components of a computer are CPU, main memory, I/O and _____.**
A. ALU
B. system interconnection ✓
C. registers
D. control unit
Answer: **B. system interconnection**
CPU, Main memory, I/O, System interconnection.
المصدر: Ch1 study guide §5

4 **In the IAS computer, which register holds the right-hand instruction of a word temporarily?**
A. MQ
B. IBR ✓
C. MBR
D. IR
Answer: **B. IBR**
IBR = Instruction Buffer Register.
المصدر: Ch1 study guide §7.1

5 **Second-generation computers differ from the IAS computer mainly by the introduction of the _____.**
A. microprocessor
B. cache
C. data channel ✓
D. integrated circuit
Answer: **C. data channel**
Data channel (IBM 7094).
المصدر: Ch1 study guide §7.2

6 **Which building block of microelectronics stores one bit of data?**
A. Interconnections
B. Memory cells ✓
C. Gates
D. Registers only
Answer: **B. Memory cells**
Gates = معالجة، Memory cells = تخزين، Interconnection = نقل.
المصدر: Ch1 study guide §7.3

7 **Which one is NOT a consequence of Moore's Law?**

A. Fewer interchip connections
B. Shorter electrical path length
C. Lower cost of logic and memory
D. Larger power and cooling requirements ✓

Answer: **D. Larger power and cooling requirements**

الصحيح: انخفاض الطاقة والتبريد.

المصدر: Ch1 study guide §7.3

8 **The Intel 4004 (1971) is considered the birth of the _____.**

A. microprocessor ✓
B. supercomputer
C. microcontroller
D. mainframe

Answer: **A. microprocessor**

أول CPU على شريحة واحدة.

المصدر: Ch1 study guide §7.4

9 **In which cloud model does the client manage ONLY the applications?**

A. PaaS ✓
B. IaaS
C. SaaS
D. Traditional IT

Answer: **A. PaaS**

PaaS: التطبيقات فقط.

المصدر: Ch1 study guide §10

10 **Which technologies help IoT objects to be identified and located?**

A. RFID
B. WiFi
C. GPS
D. All of the above ✓

Answer: **D. All of the above**

.RFID, WiFi, mobile networks, GPS

المصدر: Ch1 study guide §9

Fill in the blank — أكمل الفراغ (5)

1 **The mathematician attributed with the stored program concept is John _____.**

Answer: **von Neumann**

المصدر: Ch1 study guide §7.1

2 **The IAS main memory runs from M(0) to M(____).**

Answer: **4095**

4096 كلمة.

المصدر: Ch1 study guide §7.1

3 **An _____ system is the use of electronics and software within a product, as opposed to a general-purpose computer.**

Answer: **embedded**

المصدر: Ch1 study guide §8

4 **IaaS stands for Infrastructure as a _____.**

Answer: **Service**

المصدر: Ch1 study guide §10

5 **The accumulator register of the IAS is abbreviated _____.**

Answer: **AC**

المصدر: Ch1 study guide §7.1

Ch2: Performance Issues

True / False — (12) صح / خطأ

1 **Pipelining reduces the execution time of each individual instruction.**
Answer: **False**
يقلل زمن البرنامج الكلي لا زمن التعليمة الواحدة.
المصدر: Ch2 notes §3

2 **Superscalar execution uses multiple parallel pipelines.**
Answer: **True**
المصدر: Ch2 notes §3

3 **A superpipelined design divides pipeline stages into more granular sub-stages.**
Answer: **True**
المصدر: Ch2 notes §3

4 **Data flow analysis lets the processor create an optimized schedule by examining dependencies between instructions.**
Answer: **True**
المصدر: Ch2 notes §3

5 **An incorrect branch prediction can lead to discarded work and pipeline stalls.**
Answer: **True**
المصدر: Ch2 notes §3

6 **Speculative execution uses branch prediction and data flow analysis to execute instructions ahead of their appearance in the program.**
Answer: **True**
المصدر: Ch2 notes §3

7 **DRAM is generally much faster than the processor, so memory is never a bottleneck.**
Answer: **False**
DRAM أبطأ بكثير من المعالج.
المصدر: Ch2 notes §4

8 **Making DRAM 'wider' rather than 'deeper' increases the number of bits retrieved at one time.**
Answer: **True**
المصدر: Ch2 notes §4

9 **The overall balance of the system is more important than the raw performance of a single component.**
Answer: **True**
المصدر: Ch2 notes §4

10 **In a multicore design the number of caches decreases as cores are added.**
Answer: **False**
عدد الـ caches يزداد ولهذا ظهرت مستويات cache.
المصدر: Ch2 notes §6

11 **MIC architectures feature a large number of homogeneous general-purpose processors on a single chip.**
Answer: **True**
المصدر: Ch2 notes §7

12 **A GPU was traditionally designed for graphics processing.**
Answer: **True**
المصدر: Ch2 notes §8

Multiple Choice — (7) اختيار من متعدد

1 **Which technique focuses on issuing multiple instructions simultaneously per clock cycle?**
A. Data flow analysis
B. Superscalar execution ✓
C. Superpipelining
D. Branch prediction
Answer: **B. Superscalar execution**
المصدر: Ch2 notes §3

2 **Pipelining is best compared to _____.**
A. a faster clock only
B. a single worker doing every task
C. an assembly line ✓
D. a larger cache
Answer: **C. an assembly line**
.assembly line
المصدر: Ch2 notes §3

3 **Which is one of the three approaches to overcome the processor-memory speed imbalance?**
A. Use narrower buses
B. Increase the interconnect bandwidth ✓
C. Reduce the number of registers
D. Remove the cache
Answer: **B. Increase the interconnect bandwidth**
المصدر: Ch2 notes §4

4 **Increasing the number of bits retrieved at one time is achieved by making DRAMs _____.**
A. deeper
B. smaller
C. wider ✓
D. slower
Answer: **C. wider**
المصدر: Ch2 notes §4

5 **Which is NOT a way to improve chip organization and architecture?**

A. Graphics Processing Unit (GPU)
B. Reduce the number of gates ✓
C. Multicore
D. Increase the size and speed of caches

Answer: **B. Reduce the number of gates**

المصدر: Ch2 notes §5

6 **As caches became larger, designers created _____.**

A. only off-chip cache
B. no cache
C. two and then three levels of cache ✓
D. one level of cache

Answer: **C. two and then three levels of cache**

المصدر: Ch2 notes §6

7 **The strategy of multicore is to use _____.**

A. one very complex processor
B. two simpler processors on the chip instead of one complex processor ✓
C. only GPUs
D. a bigger clock

Answer: **B. two simpler processors on the chip instead of one complex processor**

المصدر: Ch2 notes §6

Fill in the blank — أكمل الفراغ (4)

1 **_____ analysis examines which instructions depend on each other's results to build an optimized schedule.**

Answer: **Data flow**

المصدر: Ch2 notes §3

2 **The three approaches to the memory imbalance are: wider DRAM, reduce access frequency (cache), and increase the _____ bandwidth.**

Answer: **interconnect**

المصدر: Ch2 notes §4

3 **Processor chips with a large number of homogeneous general-purpose cores are called _____ (abbreviation).**

Answer: **MIC**

.Many Integrated Core

المصدر: Ch2 notes §7

4 **Branch prediction, pipelining and superscalar execution are techniques used to speed up the _____.**

Answer: **microprocessor**

المصدر: Ch2 notes §2-3

Ch3: Function & Interconnection

True / False — (12) صح / خطأ

1 **The I/OAR register specifies a particular I/O device.**
Answer: **True**
المصدر: Ch3 notes §1

2 **The four categories of computer actions are processor-memory, processor-I/O, data processing and control.**
Answer: **True**
المصدر: Ch3 notes §2

3 **The instruction cycle state diagram has seven states.**
Answer: **True**
.if, iod, oac, of, do, os, iac
المصدر: Ch3 notes §3

4 **Operand address calculation appears twice in the instruction cycle diagram.**
Answer: **True**
مرة لجلب المعامل ومرة لتخزين النتيجة.
المصدر: Ch3 notes §3

5 **The interrupt check step is performed after the operand store.**
Answer: **True**
المصدر: Ch3 notes §4

6 **A timer interrupt is generated by an I/O controller.**
Answer: **False**
Timer interrupt من مؤقّت داخل المعالج.
المصدر: Ch3 notes §4

7 **A power failure generates a hardware failure interrupt.**
Answer: **True**
المصدر: Ch3 notes §4

8 **DMA relieves the processor of responsibility for the I/O exchange.**
Answer: **True**
المصدر: Ch3 notes §5

9 **If two devices transmit on a bus during the same time period, their signals overlap and become garbled.**
Answer: **True**
المصدر: Ch3 notes §7

10 **The width of the data bus has no effect on system performance.**
Answer: **False**
عرض الـ data bus عامل رئيسي في الأداء.
المصدر: Ch3 notes §7

11 **Address lines are generally also used to address I/O ports.**
Answer: **True**
المصدر: Ch3 notes §7

12 **The Bus grant control line indicates that a pending interrupt has been recognized.**
Answer: **False**
هذه Interrupt ACK؛ Bus grant تعني أن الـ bus مُنِح لوحدة.
المصدر: Ch3 notes §7

Multiple Choice — (8) اختيار من متعدد

1 **Which register contains the data to be written into memory or receives the data read from memory?**
A. MAR
B. MBR ✓
C. I/OAR
D. PC
Answer: **B. MBR**
المصدر: Ch3 notes §1

2 **The state that analyzes the instruction to determine the type of operation and operands is _____.**
A. instruction operation decoding ✓
B. data operation
C. operand fetch
D. instruction fetch
Answer: **A. instruction operation decoding**
.iod
المصدر: Ch3 notes §3

3 **Which state writes the result into memory or out to I/O?**
A. instruction fetch
B. data operation
C. operand store ✓
D. operand fetch
Answer: **C. operand store**
.os
المصدر: Ch3 notes §3

4 **An interrupt generated by division by zero belongs to the _____ class.**
A. program ✓
B. timer
C. I/O
D. hardware failure
Answer: **A. program**
المصدر: Ch3 notes §4

5 **Which control line indicates that a module needs to gain control of the bus?**

A. Bus request ✓
B. Transfer ACK
C. Bus grant
D. Reset

Answer: **A. Bus request**

المصدر: Ch3 notes §7

6 **Which control line is used to synchronize operations?**

A. Clock ✓
B. Reset
C. Memory read
D. Interrupt request

Answer: **A. Clock**

المصدر: Ch3 notes §7

7 **Which control line initializes all modules?**

A. Clock
B. Transfer ACK
C. Bus grant
D. Reset ✓

Answer: **D. Reset**

المصدر: Ch3 notes §7

8 **The number of lines of the data bus is called its _____.**

A. length
B. width ✓
C. speed
D. depth

Answer: **B. width**

المصدر: Ch3 notes §7

Fill in the blank — أكمل الفراغ (4)

1 **The processor-controlled sequence 'if, iod, oac, of, do, os, iac' is the instruction _____ state diagram.**

Answer: **cycle**

المصدر: Ch3 notes §3

2 **An interrupt is a signal that requests the processor to suspend its current execution and service the occurred interrupt, using an interrupt service _____.**

Answer: **routine**

.ISR

المصدر: Ch3 notes §4

3 **_____ memory access allows I/O exchanges to occur directly with memory without tying up the processor (abbreviation).**

Answer: **DMA**

.Direct Memory Access

المصدر: Ch3 notes §5

4 **The _____ line indicates that data have been accepted from or placed on the bus.**

Answer: **Transfer ACK**

المصدر: Ch3 notes §7

Ch4: Computer Memory System

True / False — (12) صح / خطأ

1 **Internal memory is accessible by the processor via an I/O module.**
Answer: **False**
هذه External memory.
المصدر: Ch4 notes §1

2 **Sequential access has a variable access time that depends on the data location.**
Answer: **True**
المصدر: Ch4 notes §2

3 **Random access has an access time that is constant, independent of prior accesses.**
Answer: **True**
المصدر: Ch4 notes §2

4 **Disk access is an example of associative access.**
Answer: **False**
القرص = Direct access.
المصدر: Ch4 notes §2

5 **Cache is an example of associative access.**
Answer: **True**
المصدر: Ch4 notes §2

6 **Going down the memory hierarchy, cost per bit decreases and access time increases.**
Answer: **True**
المصدر: Ch4 notes §3

7 **Greater capacity of memory means faster access time.**
Answer: **False**
سعة أكبر = أبطأ.
المصدر: Ch4 notes §3

8 **Registers are managed by the compiler.**
Answer: **True**
المصدر: Ch4 notes §5

9 **Main memory is typically built from SRAM.**
Answer: **False**
الذاكرة الرئيسية DRAM والـ cache SRAM.
المصدر: Ch4 notes §5

10 **A disk cache is a buffer in main memory that holds data for the disk.**
Answer: **True**
المصدر: Ch4 notes §4

11 **External nonvolatile memory is also called secondary or auxiliary memory.**
Answer: **True**
المصدر: Ch4 notes §3

12 **The unit of transfer of external memory is usually a block much larger than a word.**
Answer: **True**
المصدر: Ch4 notes §2

Multiple Choice — (6) اختيار من متعدد

1 **Tape is an example of _____ access.**
A. random
B. sequential ✓
C. direct
D. associative
Answer: **B. sequential**
المصدر: Ch4 notes §2

2 **The smallest location that can be uniquely addressed is the _____.**
A. page
B. block
C. addressable unit ✓
D. sector
Answer: **C. addressable unit**
المصدر: Ch4 notes §2

3 **The correct order of the hierarchy from fastest to slowest is _____.**
A. Cache, registers, disk, main memory
B. Registers, on-chip cache, main memory, disk ✓
C. Disk, main memory, cache, registers
D. Registers, main memory, cache, disk
Answer: **B. Registers, on-chip cache, main memory, disk**
المصدر: Ch4 notes §3

4 **The unit of transfer of a magnetic disk (secondary memory) is a _____.**
A. sector ✓
B. word
C. cache block
D. register
Answer: **A. sector**
512 bytes.
المصدر: Ch4 notes §5

5 **Which is managed by the operating system?**

A. Registers
B. Cache
C. Main memory ✓
D. None of the above

Answer: **C. Main memory**

المصدر: Ch4 notes §5

6 **The memory design questions are 'How much? How fast? How _____?'**

A. big
B. expensive ✓
C. wide
D. long

Answer: **B. expensive**

المصدر: Ch4 notes §3

Fill in the blank — أكمل الفراغ (4)

1 **_____ access is based on a portion of the word's contents rather than its address.**

Answer: **Associative**

المصدر: Ch4 notes §2

2 **The faster the access time, the _____ the cost per bit.**

Answer: **greater**

المصدر: Ch4 notes §3

3 **Off-line bulk memory typically uses magnetic _____.**

Answer: **tape**

المصدر: Ch4 notes §5

4 **_____ memory is directly accessible by the processor (registers, cache, main memory).**

Answer: **Internal**

المصدر: Ch4 notes §1

Ch5: Cache Memory

True / False — (13) صح / خطأ

1 **The cache has far fewer lines than main memory has blocks.**
Answer: **True**
المصدر: Ch5 notes §1

2 **Each cache line holds one block plus a Tag.**
Answer: **True**
المصدر: Ch5 notes §1

3 **On a cache miss, loading the block into the cache line and delivering the word to the CPU happen in parallel.**
Answer: **True**
المصدر: Ch5 notes §2

4 **A logical cache is placed between the MMU and main memory.**
Answer: **False**
هذا physical cache؛ الـ logical بين المعالج والـ MMU.
المصدر: Ch5 notes §4

5 **In direct mapping each main memory block can be loaded into only one possible cache line.**
Answer: **True**
المصدر: Ch5 notes §5.1

6 **Direct mapping needs a replacement algorithm.**
Answer: **False**
لا يوجد خيار، فلا حاجة لخوارزمية.
المصدر: Ch5 notes §6

7 **In associative mapping the cache control logic examines every line's tag simultaneously.**
Answer: **True**
المصدر: Ch5 notes §5.2

8 **In set associative mapping m = v × k, where v is the number of sets and k the lines per set.**
Answer: **True**
المصدر: Ch5 notes §5.3

9 **LRU replaces the block that has experienced the fewest references.**
Answer: **False**
هذا LFU.
المصدر: Ch5 notes §6

10 **FIFO can be implemented as a round-robin (circular buffer) technique.**
Answer: **True**
المصدر: Ch5 notes §6

11 **Write back makes all writes to both cache and main memory.**
Answer: **False**
هذا write through.
المصدر: Ch5 notes §7

12 **Write back minimizes memory writes but makes I/O access more complex.**
Answer: **True**
المصدر: Ch5 notes §7

13 **The trend is split caches at L1 and unified caches at higher levels.**
Answer: **True**
المصدر: Ch5 notes §8

Multiple Choice — (8) اختيار من متعدد

1 **With direct mapping, the cache line number is obtained by _____.**
A. block number mod m ✓
B. block number × m
C. block number + m
D. block number / m
Answer: **A. block number mod m**
المصدر: Ch5 notes §5.1

2 **Which mapping is a compromise between direct and associative mapping?**
A. Set associative ✓
B. Fully associative
C. Logical
D. Split
Answer: **A. Set associative**
المصدر: Ch5 notes §5.3

3 **Which replacement algorithm is the most popular because of its simple implementation and effectiveness?**
A. FIFO
B. Random
C. LFU
D. LRU ✓
Answer: **D. LRU**
المصدر: Ch5 notes §6

4 **The main disadvantage of write back is _____.**
A. no cache coherence issue
B. low hit rate
C. complex circuitry ✓
D. large memory traffic
Answer: **C. complex circuitry**
المصدر: Ch5 notes §7

5 **Which replacement algorithm can be implemented with a counter associated with each line?**

A. LRU
B. FIFO
C. LFU ✓
D. Random

Answer: **C. LFU**

المصدر: Ch5 notes §6

6 **L1 cache is usually _____.**

A. unified only
B. internal to the processor ✓
C. external to the processor
D. larger than L2

Answer: **B. internal to the processor**

المصدر: Ch5 notes §8

7 **The page table that translates virtual to physical addresses is managed by the _____.**

A. disk
B. cache
C. OS ✓
D. ALU

Answer: **C. OS**

المصدر: Ch5 notes §4

8 **Which of these is an element of cache design?**

A. Mapping function
B. Replacement algorithm
C. Write policy
D. All of the above ✓

Answer: **D. All of the above**

المصدر: Ch5 notes §3

Fill in the blank — أكمل الفراغ (4)

1 **In direct mapping, a block is placed in a line number computed as block number ____ m.**

Answer: **mod**

المصدر: Ch5 notes §5.1

2 **A word found in the cache is called a _____.**

Answer: **hit**

المصدر: Ch5 notes §2

3 **With _____ mapping any block can go into any line of the cache.**

Answer: **associative**

المصدر: Ch5 notes §5.2

4 **The hardware unit that translates virtual addresses to physical addresses is the _____ (abbreviation).**

Answer: **MMU**

المصدر: Ch5 notes §4

Ch6: Internal Memory

True / False — (14) صح / خطأ

1 **A DRAM cell uses one capacitor and one transistor.**
Answer: **True**
المصدر: Ch6 notes §2

2 **An SRAM cell uses six transistors.**
Answer: **True**
المصدر: Ch6 notes §3

3 **DRAM is faster and more expensive than SRAM.**
Answer: **False**
العكس.
المصدر: Ch6 notes §4

4 **SRAM is used for main memory.**
Answer: **False**
SRAM للـ cache، DRAM للذاكرة الرئيسية.
المصدر: Ch6 notes §4

5 **DRAM values must be refreshed every 10–100 ms.**
Answer: **True**
المصدر: Ch6 notes §2

6 **ROM is nonvolatile and its data is wired into the chip during manufacturing.**
Answer: **True**
المصدر: Ch6 notes §5

7 **PROM can be written only once.**
Answer: **True**
المصدر: Ch6 notes §6

8 **EPROM is erased with UV light at chip level.**
Answer: **True**
المصدر: Ch6 notes §7

9 **Flash memory provides byte-level erasure.**
Answer: **False**
.Flash block-level
المصدر: Ch6 notes §7

10 **Interleaved memory with K banks can service K requests simultaneously.**
Answer: **True**
المصدر: Ch6 notes §8

11 **A soft error causes permanent damage to the memory.**
Answer: **False**
لا يسبب ضرراً دائماً.
المصدر: Ch6 notes §9

12 **A hard failure is a permanent physical defect.**
Answer: **True**
المصدر: Ch6 notes §9

13 **DDR-SDRAM sends data twice per clock cycle.**
Answer: **True**
المصدر: Ch6 notes §11

14 **CDRAM integrates a small SRAM cache of 16 Kb on a DRAM chip.**
Answer: **True**
المصدر: Ch6 notes §11

Multiple Choice — (7) اختيار من متعدد

1 **Which memory type is volatile?**
A. ROM
B. Flash
C. RAM ✓
D. EPROM
Answer: **C. RAM**
المصدر: Ch6 notes §1

2 **Which memory type is written only once electrically by the customer?**
A. EPROM
B. ROM
C. EEPROM
D. PROM ✓
Answer: **D. PROM**
المصدر: Ch6 notes §6

3 **Which of the following is an error-correcting code method?**
A. Hamming code
B. Parity method
C. CRC
D. All of the above ✓
Answer: **D. All of the above**
المصدر: Ch6 notes §9

4 **SDRAM is synchronized with the _____.**
A. cache
B. disk
C. bus request
D. system clock ✓
Answer: **D. system clock**
المصدر: Ch6 notes §11

5 **DDR transfers data on _____.**

A. only the falling edge
B. no edge
C. both rising and falling edges ✓
D. only the rising edge

Answer: **C. both rising and falling edges**

المصدر: Ch6 notes §11

6 **Which memory is erased in blocks in a single 'flash' action?**

A. PROM
B. EPROM
C. EEPROM
D. Flash memory ✓

Answer: **D. Flash memory**

المصدر: Ch6 notes §7

7 **Which statement about SRAM is true?**

A. It requires refresh
B. It is slower than DRAM
C. It is cheaper than DRAM
D. It does not require refresh ✓

Answer: **D. It does not require refresh**

المصدر: Ch6 notes §4

Fill in the blank — أكمل الفراغ (4)

1 **DRAM stores each bit as a charge on a _____.**

Answer: **capacitor**

المصدر: Ch6 notes §2

2 **SRAM stores bits using _____ logic-gate configurations.**

Answer: **flip-flop**

المصدر: Ch6 notes §3

3 **The additional bits added to a memory word for error detection are _____ bits.**

Answer: **parity**

المصدر: Ch6 notes §9

4 **DRAM chips grouped together form a memory _____.**

Answer: **bank**

المصدر: Ch6 notes §8

Ch7: External Memory

True / False — (12) صح / خطأ

1 **Each platter has two surfaces.**
Answer: **True**
المصدر: Ch7 notes §1

2 **Tracks are concentric rings and sectors are separated by gaps.**
Answer: **True**
المصدر: Ch7 notes §1

3 **The read/write head touches the disk surface in a Winchester disk.**
Answer: **False**
الرأس يطفو على وسادة هوائية.
المصدر: Ch7 notes §2

4 **A fixed-head disk has one read/write head per track.**
Answer: **True**
المصدر: Ch7 notes §2

5 **A movable-head disk has one head per surface.**
Answer: **True**
المصدر: Ch7 notes §2

6 **A narrower head allows narrower tracks and greater data density.**
Answer: **True**
المصدر: Ch7 notes §2

7 **Glass substrate improves shock resistance and reduces surface defects.**
Answer: **True**
المصدر: Ch7 notes §1

8 **An SSD has moving mechanical parts.**
Answer: **False**
المصدر: Ch7 notes §3

9 **An SSD generally consumes less power than an HDD.**
Answer: **True**
المصدر: Ch7 notes §3

10 **SSD shock resistance is up to 1500g.**
Answer: **True**
المصدر: Ch7 notes §3

11 **CD capacity is about 700 MB and Blu-ray capacity ranges from 25 to 128 GB.**
Answer: **True**
المصدر: Ch7 notes §5

12 **CD-R can be written multiple times.**
Answer: **False**
مرة واحدة فقط.
المصدر: Ch7 notes §5

Multiple Choice — (6) اختيار من متعدد

1 **Which is NOT a head mechanism?**
A. Aerodynamic gap
B. Sealed gap ✓
C. Contact
D. Fixed gap
Answer: **B. Sealed gap**
المصدر: Ch7 notes §2

2 **A removable disk provides _____.**
A. a fixed capacity
B. unlimited storage capacity ✓
C. no data transfer
D. less portability
Answer: **B. unlimited storage capacity**
المصدر: Ch7 notes §2

3 **The SSD controller is responsible for _____.**
A. spinning the platters
B. managing the SSD ✓
C. generating light
D. holding the data
Answer: **B. managing the SSD**
المصدر: Ch7 notes §4

4 **Which is part of an SSD?**
A. Platter
B. Spindle
C. NAND flash memory ✓
D. Actuator arm
Answer: **C. NAND flash memory**
المصدر: Ch7 notes §3

5 **Which optical disc has the largest capacity?**
A. Blu-ray ✓
B. CD-ROM
C. CD
D. DVD
Answer: **A. Blu-ray**
المصدر: Ch7 notes §5

6 **Which optical disc type can be recorded once?**
A. ROM
B. R ✓
C. RW
D. None
Answer: **B. R**
المصدر: Ch7 notes §5

Fill in the blank — (4) أكمل الفراغ

1 **Disks rotate around a central _____.**

Answer: **spindle**

المصدر: Ch7 notes §1

2 **The head mechanism used in sealed drives that fly closest to the surface is the _____ gap (Winchester).**

Answer: **aerodynamic**

المصدر: Ch7 notes §2

3 **A single-sided disk has the magnetic coating on _____ side only.**

Answer: **one**

المصدر: Ch7 notes §2

4 **The abbreviation SSD stands for _____ State Drive.**

Answer: **Solid**

المصدر: Ch7 notes §3

Ch14: Addressing Modes & Formats

True / False — (13) صح / خطأ

1 **In immediate addressing the operand is contained in the instruction.**
Answer: **True**
Operand = A
المصدر: Ch14 notes §2

2 **Direct addressing requires two memory references.**
Answer: **False**
مرجع ذاكرة واحد.
المصدر: Ch14 notes §3

3 **In indirect addressing, EA = (A).**
Answer: **True**
المصدر: Ch14 notes §4

4 **In register addressing the operand is in a register.**
Answer: **True**
المصدر: Ch14 notes §5

5 **In register indirect addressing, EA = (R).**
Answer: **True**
المصدر: Ch14 notes §6

6 **Relative addressing computes EA = A + (PC).**
Answer: **True**
المصدر: Ch14 notes §7.1

7 **In base-register addressing the register holds the displacement and the address field holds the main memory address.**
Answer: **False**
العكس: هذا indexing.
المصدر: Ch14 notes §7.2-7.3

8 **Indexing is an efficient mechanism for loops.**
Answer: **True**
المصدر: Ch14 notes §7.3

9 **Stack addressing is a form of implied addressing.**
Answer: **True**
المصدر: Ch14 notes §8

10 **Immediate addressing is used to implement pointers.**
Answer: **False**
المؤشرات = indirect / register indirect.
المصدر: Ch14 notes §9

11 **Relative addressing is useful for branch instructions.**
Answer: **True**
المصدر: Ch14 notes §9

12 **A mode field is used to tell the processor which addressing mode is used.**
Answer: **True**
المصدر: Ch14 notes §10

13 **Variable-length instructions increase the complexity of the processor.**
Answer: **True**
المصدر: Ch14 notes §13

Multiple Choice — (8) اختيار من متعدد

1 **Which addressing mode is used to initialize registers to a constant value?**
A. Indirect
B. Immediate ✓
C. Direct
D. Indexing
Answer: **B. Immediate**
المصدر: Ch14 notes §9

2 **Which addressing mode is used for recursive procedures?**
A. Immediate
B. Direct
C. Relative
D. Base-register ✓
Answer: **D. Base-register**
المصدر: Ch14 notes §9

3 **Which addressing mode is used for arrays and records?**
A. Register
B. Stack
C. Immediate
D. Indexing ✓
Answer: **D. Indexing**
المصدر: Ch14 notes §9

4 **EA = A + (PC) belongs to _____ addressing.**
A. stack
B. indirect
C. direct
D. relative ✓
Answer: **D. relative**
المصدر: Ch14 notes §7.1

5 **Which mode needs two memory references to get the operand?**

A. Immediate
B. Indirect ✓
C. Register
D. Direct

Answer: **B. Indirect**

المصدر: Ch14 notes §4

6 **Which of the following is NOT a factor in allocating the addressing bits?**

A. Number of addressing modes
B. Number of operands
C. The screen resolution ✓
D. Address granularity

Answer: **C. The screen resolution**

المصدر: Ch14 notes §12

7 **The instruction length should be equal to the memory-transfer length or _____.**

A. unrelated
B. half of it
C. a multiple of it ✓
D. random

Answer: **C. a multiple of it**

المصدر: Ch14 notes §11

8 **Which mode combines direct and register indirect capabilities?**

A. Stack
B. Displacement ✓
C. Immediate
D. Register

Answer: **B. Displacement**

المصدر: Ch14 notes §7

Fill in the blank — أكمل الفراغ (4)

1 **EA stands for _____ address.**

Answer: **Effective**

المصدر: Ch14 notes §1

2 **In stack addressing the top of the stack is pointed to by the stack _____.**

Answer: **pointer**

المصدر: Ch14 notes §8

3 **The displacement addressing types are relative, base-register and _____.**

Answer: **indexing**

المصدر: Ch14 notes §7

4 **A stack is a _____ (LIFO) structure.**

Answer: **last-in-first-out**

المصدر: Ch14 notes §8

Ch16: Processor Structure & Function

True / False — (13) صح / خطأ

1 **The ALU contains status flags, a shifter, a complementer and arithmetic and Boolean logic.**
Answer: **True**
المصدر: Ch16 notes §1

2 **User-visible registers allow the programmer to minimize main memory references.**
Answer: **True**
المصدر: Ch16 notes §2

3 **The zero flag is set when the result of the last operation is zero.**
Answer: **True**
المصدر: Ch16 notes §2

4 **During the fetch cycle the PC value is moved to the MBR first.**
Answer: **False**
PC → MAR أولاً.
المصدر: Ch16 notes §3.1

5 **The PC is incremented during the fetch cycle.**
Answer: **True**
المصدر: Ch16 notes §3.1

6 **In the interrupt cycle, the PC is loaded with the address of the interrupt routine.**
Answer: **True**
المصدر: Ch16 notes §3.3

7 **Pipelining performance is limited by unequal stage durations, conditional branches and interrupts.**
Answer: **True**
المصدر: Ch16 notes §4

8 **A resource hazard occurs when two instructions need the same resource.**
Answer: **True**
المصدر: Ch16 notes §5.1

9 **RAW is called an antidependency.**
Answer: **False**
.RAW = true dependency، WAR = antidependency
المصدر: Ch16 notes §5.2

10 **WAW is called output dependency.**
Answer: **True**
المصدر: Ch16 notes §5.2

11 **A loop buffer stores loop instructions in a small fast memory.**
Answer: **True**
المصدر: Ch16 notes §5.3

12 **Delayed branch executes the instruction after the branch (delay slot) before the branch takes effect.**
Answer: **True**
المصدر: Ch16 notes §5.3

13 **The multiple streams technique requires less hardware than other branch methods.**
Answer: **False**
يتطلب عتاداً أكثر.
المصدر: Ch16 notes §5.3

Multiple Choice — (8) اختيار من متعدد

1 **Which flag is set when an operation produces a carry out of the leftmost bit?**
A. Equal
B. Zero
C. Sign
D. Carry ✓
Answer: **D. Carry**
المصدر: Ch16 notes §2

2 **In the fetch cycle the instruction read from memory is first held in the _____ and then moved to the IR.**
A. PC
B. MBR ✓
C. MAR
D. ALU
Answer: **B. MBR**
المصدر: Ch16 notes §3.1

3 **In the indirect cycle the right-most N bits of the MBR are transferred to the _____.**
A. IR
B. PC
C. MAR ✓
D. ALU
Answer: **C. MAR**
المصدر: Ch16 notes §3.2

4 **In the interrupt cycle the saved PC is written to _____.**
A. the disk
B. the IR
C. the ALU
D. a reserved memory location (e.g. stack pointer) ✓
Answer: **D. a reserved memory location (e.g. stack pointer)**
المصدر: Ch16 notes §3.3

5 **Which hazard occurs when the pipeline makes the wrong branch prediction decision?**

A. Resource hazard
B. Data hazard
C. Control hazard ✓
D. None

Answer: **C. Control hazard**

المصدر: Ch16 notes §5.3

6 **Which data hazard happens when the read takes place before the write operation is complete?**

A. RAW ✓
B. WAR
C. WAW
D. None

Answer: **A. RAW**

المصدر: Ch16 notes §5.2

7 **Which technique fetches instructions from more than one possible path?**

A. Multiple streams ✓
B. Delayed branch
C. Prefetch branch target
D. Loop buffer

Answer: **A. Multiple streams**

المصدر: Ch16 notes §5.3

8 **Which of the following is another name for a pipeline hazard?**

A. Pipeline cache
B. Pipeline flag
C. Pipeline bus
D. Pipeline bubble ✓

Answer: **D. Pipeline bubble**

المصدر: Ch16 notes §5

Fill in the blank — أكمل الفراغ (4)

1 **The PSW flag that shows whether the result was positive or negative is the _____ flag.**

Answer: **Sign**

المصدر: Ch16 notes §2

2 **WAR stands for Write After _____.**

Answer: **Read**

المصدر: Ch16 notes §5.2

3 **The instruction register holds the instruction that was _____.**

Answer: **fetched**

المصدر: Ch16 notes §3

4 **The five techniques for conditional branches are multiple streams, prefetch branch target, loop buffer, branch prediction and _____ branch.**

Answer: **delayed**

المصدر: Ch16 notes §5.3